GenAI Medical Report Generator

Generative AI 2

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Introduction

What is Generative AI?

AI that creates new content by learning patterns from existing data

AI in Healthcare

Transforming diagnosis, treatment planning, and patient care through intelligent automation

The Problem

Manual medical reports are time-consuming and prone to human error

Our Solution

Automated report generation using AI to save time and improve accuracy

Project Overview

01

Web-Based Application

Accessible through any modern web browser

02

Patient Details Input

Collect comprehensive patient information through structured forms

03

SOAP Report Generation

Creates structured reports using Subjective, Objective, Assessment, Plan format

04

ICD-10 Suggestions

Provides standardized medical coding recommendations

05

Safety Disclaimer

Includes appropriate medical disclaimers for responsible usage

Technologies Used

Frontend: Streamlit

Python-based framework for building interactive user interfaces

Backend: FastAPI

Modern Python framework for building APIs with automatic documentation

AI Model: Transformer-based

FLAN-T5 or GPT-type model for natural language generation

Data Format: JSON

Lightweight data interchange format for structured data transfer

Server: Uvicorn

ASGI server for high-performance Python web applications

Working Flow

The system follows a streamlined workflow from data input to report generation, ensuring efficiency and accuracy at each step.

Features & Advantages

1

Fast Report Generation

Creates comprehensive medical reports in seconds instead of minutes

2

Easy-to-Use Interface

Intuitive design requiring minimal training for healthcare professionals

3

AI-Based Dynamic Output

Adapts to different patient scenarios with context-aware generation

4

Structured SOAP Format

Follows standardized medical documentation practices

5

Scalable Architecture

Can handle increasing patient load with cloud-based infrastructure

Challenges & Future Scope

Current Challenges

Model Dependency

Output accuracy depends on training data quality and model capabilities

Limited Validation

Requires extensive medical validation and testing

Future Improvements

1. **Advanced LLMs**
   Integration of more sophisticated language models
2. **PDF Export**
   Direct download capability for generated reports
3. **Medical Databases**
   Integration with real patient databases and EHR systems

Key Takeaways

Time Efficiency

Reduces report generation time from minutes to seconds

Standardization

Ensures consistent SOAP format across all reports

Extendable Design

Architecture supports future enhancements and integrations

Thank You

Any Questions?

Project Screenshots

[code]

[GUI]

[Generated report]

[Backend]